MADHU CHANDRIKA BALRAM

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SUMMARY

Software Engineer with 4+ years of full-stack and backend experience specializing in building high-throughput microservices, scalable distributed systems, and production GenAI software. Proven track record in designing resilient FastAPI and Spring Boot architectures, event-driven data pipelines, and multi-agent RAG workflows across enterprise cloud environments. Adept at applying core software engineering practices—object-oriented design, and automated testing—to build secure, enterprise-grade AI applications.

SKILLS

Core Software & Systems: Python, Java, C++, TypeScript, JavaScript, SQL, REST APIs, Microservices Architecture, Event-Driven Architecture, System Design, Design Patterns, Object-Oriented Design, Data Structures, Algorithms
Backend & Web Engineering: FastAPI, Spring Boot, Node.js, React, Next.js, gRPC, Multithreading, Concurrency, AsyncIO, Kafka, RabbitMQ, OAuth 2.0, JWT, Microsoft Entra ID, Role-Based Access Control
AI, GenAI & Agentic Systems: Generative AI, Large Language Models, RAG Architecture, AI Agents, Agentic Workflows, Multi-Agent Systems, LangChain, LangGraph, LlamaIndex, OpenAI APIs, Azure OpenAI, Model Fine-Tuning, vLLM, Ragas, LangSmith
Databases & Data Processing: Azure AI Search, Pinecone, Milvus, Qdrant, ChromaDB, pgvector, PostgreSQL, MySQL, MongoDB, Azure Cosmos DB, Redis, Elasticsearch, Apache Spark, Data Pipelines
Cloud Services: AWS EC2, AWS S3, AWS SageMaker, AWS Lambda, AWS RDS, AWS EKS, Azure App Service, Azure Static Web Apps, GCP
DevOps & MLOps: Docker, Kubernetes, Terraform, Git, CI CD Pipelines, GitHub Actions, Azure DevOps, Grafana, MLflow, TIS
Software Quality & Testing: PyTest, JUnit, Mockito, Spring Test, Jest, Postman, Unit Testing, Integration Testing

PROFESSIONAL EXPERIENCE

GEP, USA | *AI Software Engineer* **January 2026 – Present**

- Architected autonomous multi-agent RAG workflows using LangChain, LangGraph, Python, and Azure OpenAI for contract analysis and clause extraction, incorporating tool orchestration, guardrails, and RAG evaluation for groundedness and retrieval quality.
- Engineered high-throughput FastAPI backend microservices in Python deployed on Azure App Service, integrated with Azure AI Search for hybrid retrieval, Redis, and pgvector for embedding storage and similarity search, serving 100K+ daily API requests at 99.9% uptime.
- Built concurrent document-processing pipelines using Python AsyncIO, Azure Cosmos DB, and Azure AI Search indexing, implementing automated regression testing to accelerate document indexing speeds by 35%.
- Implemented enterprise security frameworks using OAuth 2.0, JWT, Microsoft Entra ID, and role-based access control (RBAC) across distributed microservices, safeguarding corporate procurement data.
- Deployed containerized vLLM model-serving infrastructure on Azure Kubernetes Service (AKS) using Docker and Azure DevOps CI CD to cut inference costs by 25%, building TypeScript, React, and Next.js dashboards to track agent execution telemetry.

Accenture, India | *Associate Software Engineer* **December 2022 – July 2024**

- Refactored legacy Java & Spring Boot backend services and optimized complex SQL queries on AWS RDS PostgreSQL, reducing database query execution times and decreasing recurring production defects by 25%.
- Diagnosed critical multi-tier application failures across EC2-hosted monoliths and AWS Lambda serverless functions by tracing REST API payloads, Java application logs, and AWS S3 event notification triggers.
- Authored comprehensive Unit and Integration test suites using JUnit, Mockito, and Spring Test, integrating them into Jenkins CI pipelines to achieve 80%+ code coverage and prevent regressions.
- Implemented request DTO validation using Jakarta Bean Validation across Spring Boot REST APIs, significantly reducing validation-related failures and malformed payload processing errors across release cycles.
- Engineered event-driven microservices using Java, Spring Boot, and Apache Kafka, containerizing applications with Docker for deployment on Kubernetes clusters on AWS EKS.

Mindtree, India | *Software Engineer* **January 2021 – November 2022**

- Engineered high-concurrency backend execution modules in java, optimizing low-level memory allocation routines, JVM memory usage, garbage-collection (GC) tuning, and thread-pool sizing to reduce core transaction processing latencies by 30%.
- Developed full-stack web applications using React, TypeScript, and Node.js REST APIs connected to PostgreSQL databases, improving client-side dashboard page load speeds by 35%.
- Automated CI CD deployment pipelines using Jenkins, Git, Docker, and NGINX as a reverse proxy, automating build validation tasks and increasing weekly release frequencies by 40%.
- Implemented OAuth 2.0 authorization and role-based access control (RBAC) across Node.js microservices to protect API access and prevent unauthorized endpoint execution.
- Constructed automated unit and integration test suites using JUnit and Mockito for Java components, Jest for React and Node.js modules, and Postman for REST API verification, achieving over 85% test coverage.

EDUCATION

MS in Computer Science, *DePaul University, Chicago,* **September 2024 – June 2026**
Bachelor of Technology in Electronics and Communication Engineering, *Reva University, Bangalore, India* **August 2018 – July 2022**

PROJECTS

OptiFi | AI-Powered Personal Finance Platform | *FastAPI, Next.js, PostgreSQL, SQLAlchemy, React, TypeScript, Pandas, Groq API*

- Architected a low-latency financial analytics engine using FastAPI, Next.js, and PostgreSQL, leveraging Pandas and SQLAlchemy to ingest transaction CSVs and route context-aware user queries for sub-200ms Groq API recommendations.

gRPC-Based File Retrieval Service | *Java, gRPC, Protocol Buffers, Multithreading*

- Engineered a high-performance Java gRPC search engine using Protocol Buffers and client-side multi-threading to construct a thread-safe server inverted index for fast, ranked document retrieval across distributed text files.